1 Introduction

Matroid bandits are a canonical model for stochastic learning with combinatorial feasibility. A known matroid $M = (E, \mathcal{I})$ specifies which sets of elements can be played, each element $e \in E$ is a stochastic arm with unknown mean μ_e, the reward of a basis is additive, and semi-bandit feedback reveals the individual rewards of the selected elements. This model is attractive because the statistical and combinatorial structures align: if the mean vector were known, the optimal action would be a maximum-weight basis, which is found by the greedy algorithm; when the means are unknown, optimism and concentration can be combined with matroid exchange to compare bases element by element.

The classical matroid-bandit literature is built on a fixed ground set. Kveton et al. [8] introduced optimistic matroid maximization and proved logarithmic gap-dependent regret and $O(\sqrt{KLT \log T})$ gap-free regret, where L is the ground-set size and K is the rank. Subsequent work developed tight combinatorial semi-bandit bounds [9], polymatroid semi-bandits [10], KL-based matroid algorithms [18], Thompson sampling [21], and computationally efficient matroid semi-bandit methods [15, 19]. In all these models, every element remains available for the whole horizon. An element that is uncertain today can be sampled tomorrow, and the learner can maintain arm-specific statistics for the full ground set.

This paper asks what happens when that availability assumption is removed while the matroid-bandit objective is kept. The learner starts with an initial basis, the remaining elements arrive in a single pass, at most m arms can be stored, and every played action must be a basis of the matroid restricted to the currently stored arms. Once an arriving arm is discarded, both the arm and its future samples are lost forever. The learner therefore faces a problem that is absent from standard matroid bandits: before the statistical evidence is fully reliable, it must decide which elements deserve permanent memory slots.

From the matroid point of view, this is an exchange-driven screening problem. When a new element e arrives and the learner holds a reference basis B, the only ways to insert e into a basis are to remove elements from the fundamental circuit $C_B(e)$. Thus the question "should e be kept?" is really a local basis-exchange question: is e good enough to replace one of the elements it can exchange with, and can this be decided using only the samples collected before e disappears from the stream? The cost of answering this question is the central phenomenon in the paper.

Memory restrictions have been studied for ordinary multi-armed bandits. In constant-space stochastic bandits, the learner cannot maintain statistics for all arms [13, 16]. A stricter streaming arm-memory model assumes that arms arrive online, only stored arms can be pulled, and discarded arms are lost forever. Maiti et al. [14] developed bounded-memory guarantees and identified a $T^{2/3}$ obstruction for single-pass regret minimization. Agarwal et al. [1] characterized multi-pass memory-regret tradeoffs, and Wang [20] gave tight single-pass regret bounds. These are rank-one models. The present work studies the same streaming retention bottleneck under matroid basis feasibility and semi-bandit feedback. Ordinary stochastic bandits appear as the rank-one endpoint, where the sharp gap-dependent and minimax theories are classical [3, 4, 5, 6, 11, 12, 17].

The following table places the model relative to nearby lines of work. For previous fixed-ground-set semi-bandits, L denotes the number of items, K the maximum action size or rank, and N the horizon. In the rank-one streaming row, K denotes the number of streamed arms as in that literature. For this paper, the notation is that of the rest of the paper: n arms, rank r, horizon T, stream parameter $\kappa = n - r + 1$, and global basis gap Δ_B.

Model and references	Availability and memory	Gap-free regret	Gap-dependent regret
Standard matroid bandits [8, 18, 21]	All elements remain available and their statistics can be stored; a greedy oracle returns a maximum-weight basis.	OMM gives $O(\sqrt{KLN\log N})$; partition-matroid lower bounds give $\Omega(\min\{\sqrt{KLN}, KN\})$ up to logarithmic factors [10].	OMM gives logarithmic bounds in exchange gaps, of the form $\sum_{e\notin B^*}\sum_k O((\log N)/\Delta_{e,k})$. Thompson sampling has the same logarithmic scaling up to lower-order terms, and KL-OSM matches the Lai–Robbins-type matroid lower bound asymptotically for Bernoulli rewards.
General combinatorial and polymatroid semi-bandits [7, 9, 10]	All items remain available; feasible actions are oracle-defined sets or greedy polymatroid actions.	Upper $O(\sqrt{KLN\log N})$ and lower $\Omega(\sqrt{KLN})$ in hard combinatorial families; polymatroid lower bounds give $\Omega(\min\{\sqrt{KLN}, KN\})$.	Worst-case families have $O(KL\log N/\Delta)$-type upper bounds with matching lower bounds; polymatroid UCB obtains $O(L\log N/\Delta)$ under its item gap.
Streaming arm-memory bandits [1, 14, 20]	Rank-one actions; arms arrive in a stream, memory stores only a few arms, and discarded arms are lost.	Tight single-pass rank-one rates include the characteristic $\Theta(K^{1/3}T^{2/3})$ memory-limited term, up to model-dependent memory parameters.	Gap-dependent rates include an unavoidable streaming screening cost before the best arm can be retained with high confidence; the rank-one analogue is of order $\tilde{O}(K/\Delta^2)$ with matching lower-bound behavior up to logarithmic factors.
This paper	Matroid bases must remain feasible throughout a single-pass stream; memory is at most m arms and discarded elements cannot be recovered.	Upper $O(r\kappa^{1/3}T^{2/3}\log^{1/3}(nT))$; lower $\Omega(r(n-r)^{1/3}T^{2/3})$ in the small-memory regime.	Upper $O(r\kappa\log(nT)/\Delta_B^2)$; lower $\Omega(r\min\{(n-r)/\Delta_B^2, \Delta_B T\})$, and $\Omega(r(n-r)\log T/\Delta_B^2)$ for consistent algorithms.

Our main contribution is the first regret theory for single-pass streaming matroid semi-bandits with a running basis-feasibility constraint. The algorithm keeps a reference basis and tests each arriving element only through the fundamental circuit that determines its possible exchanges. The reference basis is updated as an empirical maximum-weight basis over the portion of the stream seen so far; after the stream ends, the algorithm runs UCB on the retained memory. This gives a gap-free regret bound of order $r\kappa^{1/3}T^{2/3}\log^{1/3}(nT)$ and, when the optimal basis is unique with global basis gap Δ_B, a gap-dependent bound of order $r\kappa\log(nT)/\Delta_B^2$.

The lower bounds show that the screening cost is intrinsic. Direct sums of rank-one partition matroids force $\Omega(r(n-r)^{1/3}T^{2/3})$ gap-free regret in the small-memory regime. For fixed gap, they force $\Omega(r\min\{(n-r)/\Delta_B^2, \Delta_B T\})$ regret for arbitrary algorithms and $\Omega(r(n-r)\log T/\Delta_B^2)$ regret for consistent algorithms. Thus, up to logarithmic factors and constants, the price of streaming availability is exactly the cost of identifying enough exchange-worthy elements before they leave memory.

2 Model

Let $M = (E, \mathcal{I})$ be a loopless rank-r matroid on $E = \{1, \dots, n\}$, and let $\mathcal{B}$ be its family of bases. At each round t, a reward vector $X_t \in [0,1]^E$ is drawn independently across rounds and arms. Write $\mu_e = \mathbb{E}[X_t(e)]$ and $\mu(S) = \sum_{e \in S} \mu_e$.

The learner is given an initial basis $B_0 \in \mathcal{B}$. The other arms arrive in a fixed single-pass order

$$\pi = (\pi_1, \dots, \pi_{n-r}), \qquad \{\pi_1, \dots, \pi_{n-r}\} = E \setminus B_0.$$

The memory budget satisfies $r + 1 \le m < n$. Let H_t be the memory before the play step of round t. The running-feasibility constraint is

$$|H_t| \le m, \qquad \mathrm{rank}_M(H_t) = r.$$

Thus the learner can always choose a basis $S_t \in \mathcal{B}(M|H_t)$ and receives semi-bandit feedback $\{X_t(e) : e \in S_t\}$.

Let

$$B^\star \in \arg\max_{B \in \mathcal{B}} \mu(B), \qquad \mathrm{OPT} = \mu(B^\star).$$

The expected regret is

$$R_T = \mathbb{E}\left[\sum_{t=1}^{T} \big(\mathrm{OPT} - \mu(S_t)\big)\right].$$

3 Algorithm

The algorithm maintains a reference basis $B_{\mathrm{ref}} \subseteq H$. During the streaming phase, each arriving arm is sampled L_1 times through its fundamental circuit with respect to B_{ref}. Let L_2 be the length of the final UCB phase, so the horizon is

$$T = (n - r + 1)L_1 + L_2.$$

The extra L_1 block samples the initial basis. Screening feedback from arms other than the newly arrived arm is ignored in the analysis; using it can only improve empirical estimates.

Algorithm 1 Streaming Basis Sampling

Require: Initial basis B_0, stream $\pi_1, \dots, \pi_{n-r}$, memory budget m, screening length L_1, UCB length L_2

Ensure: Final memory H_{fin} and UCB-phase bases

```
H ← B_0, B_ref ← B_0
for all e ∈ B_0 do
    N_e ← 0, μ̂_e ← 0
end for

Sample the initial basis.
for s = 1, ..., L_1 do
    Play B_0 and observe X_s(e) for all e ∈ B_0
end for
for all e ∈ B_0 do
    μ̂_e ← L_1^{-1} Σ_{s=1}^{L_1} X_s(e), N_e ← L_1
end for
```

Process the stream.
for $i = 1, \ldots, n-r$ **do**
  $e \leftarrow \pi_i$
  if $|H| = m$ **then**
    Choose $d \in \arg\min_{a \in H \setminus B_{\text{ref}}} \widehat{\mu}_a$ and set $H \leftarrow H \setminus \{d\}$
  end if
  $H \leftarrow H \cup \{e\}$, $N_e \leftarrow 0$, $\widehat{\mu}_e \leftarrow 0$
  Choose $b_e \in \arg\min_{b \in C_{B_{\text{ref}}}(e) \setminus \{e\}} \widehat{\mu}_b$
  for $s = 1, \ldots, L_1$ **do**
    Play $B_{\text{ref}} - b_e + e$ and record $X_s(e)$
  end for
  $\widehat{\mu}_e \leftarrow L_1^{-1} \sum_{s=1}^{L_1} X_s(e)$, $N_e \leftarrow L_1$
  if $\widehat{\mu}_e > \widehat{\mu}_{b_e}$ **then**
    $B_{\text{ref}} \leftarrow B_{\text{ref}} - b_e + e$
  end if
end for

Run UCB on the final memory.
$H_{\text{fin}} \leftarrow H$, $\beta \leftarrow \log(4nT^2)$
for $t = 1, \ldots, L_2$ **do**
  for all $e \in H_{\text{fin}}$ **do**
    $U_t(e) \leftarrow \widehat{\mu}_e + \sqrt{\beta/(2N_e)}$
  end for
  Choose $S_t \in \arg\max_{S \in \mathcal{B}(M|H_{\text{fin}})} \sum_{e \in S} U_t(e)$
  Play S_t and observe $\{X_t(e) : e \in S_t\}$
  for all $e \in S_t$ **do**
    $N_e \leftarrow N_e + 1$
    $\widehat{\mu}_e \leftarrow \big((N_e - 1)\widehat{\mu}_e + X_t(e)\big)/N_e$
  end for
end for

4 Regret Analysis

Set

$$\kappa = n - r + 1, \qquad a_T = \log(4nT^2).$$

The term κ counts the $n-r$ stream arms plus the initial-basis sampling block.

Lemma 4.1 (Basis gap along exchanges) *Assume that $B^\star$ is the unique optimal basis and has global basis gap $\Delta_B > 0$. Then for every basis B,*

$$\mu(B^\star) - \mu(B) \geq |B^\star \setminus B| \, \Delta_B.$$

Proof: Let $D = B^\star \setminus B$ and $A = B \setminus B^\star$. By the strong basis exchange theorem, there is a bijection $\phi : D \to A$ such that $B^\star - e + \phi(e)$ is a basis for every $e \in D$. This basis is different from $B^\star$, hence

$$\mu_e - \mu_{\phi(e)} = \mu(B^\star) - \mu(B^\star - e + \phi(e)) \geq \Delta_B.$$

Summing over $e \in D$ gives the claim. □

Theorem 4.2 (Gap-free regret bound) *Choose*

$$L_1 = \left\lceil 2^{-1/3} \left(\frac{T^2 a_T}{\kappa^2} \right)^{1/3} \right\rceil, \qquad L_2 = T - \kappa L_1,$$

and assume T is large enough that $L_2 > 0$. Then Algorithm 1 satisfies

$$R_T \le \frac{3}{2^{1/3}} r \kappa^{1/3} T^{2/3} a_T^{1/3} + 2\sqrt{2a_T}\, m \left(\sqrt{L_1 + \frac{rL_2}{m}} - \sqrt{L_1} \right) + r\kappa + r.$$

In particular, for fixed n, m, r, $R_T = o(T)$.

The theorem can be read in the following simpler order form. Since $a_T = \Theta(\log(nT))$ and

$$m \left(\sqrt{L_1 + \frac{rL_2}{m}} - \sqrt{L_1} \right) \le \frac{rL_2}{2\sqrt{L_1}},$$

the UCB term is also of order $r\kappa^{1/3} T^{2/3} a_T^{1/3}$. Moreover, under $L_2 > 0$ we have $T > \kappa$ and hence $r\kappa + r$ is absorbed by the same leading order. Thus

$$R_T = O\left(r\kappa^{1/3} T^{2/3} \log^{1/3}(nT) \right).$$

The separated bound above is still useful because it shows the individual screening, memory-loss, and final-UCB contributions. In the coarser asymptotic order, however, all three terms are absorbed into the single $T^{2/3} \log^{1/3} T$ rate. For fixed n, m, r, the average regret is $O(T^{-1/3} \log^{1/3} T)$.

Proof: The streaming phase lasts κL_1 rounds. Since each basis has r arms and rewards lie in $[0, 1]$, each round contributes at most r regret:

$$R_{\text{stream}} \le r\kappa L_1.$$

We next record the basis invariant used to bound the memory loss. Let

$$F_i = B_0 \cup \{\pi_1, \ldots, \pi_i\}.$$

After processing π_i, the reference basis is an empirical maximum-weight basis in $M|F_i$:

$$B_{\text{ref}} \in \operatorname*{arg\,max}_{B \in \mathcal{B}(M|F_i)} \widehat{\mu}(B), \qquad \widehat{\mu}(B) = \sum_{e \in B} \widehat{\mu}_e.$$

This holds by induction. It is trivial for $i = 0$. Suppose it holds for $i - 1$, and let $e = \pi_i$. For any basis A of $M|F_i$ containing e, the strong symmetric basis exchange theorem applied to the two bases A and B_{ref} in $M|F_i$, and to the element $e \in A \setminus B_{\text{ref}}$, gives an element $f \in B_{\text{ref}} \setminus A$ such that both $A - e + f$ and $B_{\text{ref}} - f + e$ are bases. The second condition implies $f \in C_{B_{\text{ref}}}(e) \setminus \{e\}$, and the first basis lies in $M|F_{i-1}$. Hence

$$\widehat{\mu}(A) - \widehat{\mu}_e + \widehat{\mu}_f \le \widehat{\mu}(B_{\text{ref}}).$$

Since b_e is chosen to minimize $\widehat{\mu}_b$ over $C_{B_{\text{ref}}}(e) \setminus \{e\}$, this element f satisfies $\widehat{\mu}_f \ge \widehat{\mu}_{b_e}$. If $\widehat{\mu}_e \le \widehat{\mu}_{b_e}$, then $\widehat{\mu}_f \ge \widehat{\mu}_{b_e} \ge \widehat{\mu}_e$, so $\widehat{\mu}(A) \le \widehat{\mu}(B_{\text{ref}})$. If $\widehat{\mu}_e > \widehat{\mu}_{b_e}$, set

$$B' = B_{\text{ref}} - b_e + e.$$

This is a basis because $b_e \in C_{B_{\mathrm{ref}}}(e) \setminus \{e\}$. Moreover,

$$\widehat{\mu}(B') = \widehat{\mu}(B_{\mathrm{ref}}) + \widehat{\mu}_e - \widehat{\mu}_{b_e}.$$

For the arbitrary basis A containing e, the preceding inequality gives

$$\widehat{\mu}(A) \le \widehat{\mu}(B_{\mathrm{ref}}) + \widehat{\mu}_e - \widehat{\mu}_f \le \widehat{\mu}(B_{\mathrm{ref}}) + \widehat{\mu}_e - \widehat{\mu}_{b_e} = \widehat{\mu}(B').$$

Thus the updated basis B' is at least as good as every basis containing e. Bases not containing e are handled by the induction hypothesis; in the second case they are also dominated by B' because $\widehat{\mu}(B') > \widehat{\mu}(B_{\mathrm{ref}})$.

Thus after the stream ends, B_{ref} is an empirical maximum-weight basis of the full matroid. Define

$$B_H^\star \in \operatorname*{arg\,max}_{B \in \mathcal{B}(M|H_{\mathrm{fin}})} \mu(B), \qquad \mathrm{OPT}_H = \mu(B_H^\star),$$

and the memory loss

$$\Lambda = \mathrm{OPT} - \mathrm{OPT}_H .$$

Since $B_{\mathrm{ref}} \subseteq H_{\mathrm{fin}}$, $\Lambda \le \mathrm{OPT} - \mu(B_{\mathrm{ref}})$.

Let

$$\mathcal{E}_1 = \{\forall e \in E : |\widehat{\mu}_e - \mu_e| \le \varepsilon\}, \qquad \varepsilon = \sqrt{\frac{a_T}{2L_1}}.$$

Hoeffding's inequality and a union bound give $\Pr(\mathcal{E}_1^c) \le 2ne^{-a_T} \le 1/(2T^2)$. On $\mathcal{E}_1$, every basis estimate is within $r\varepsilon$ of its mean, and empirical optimality of B_{ref} gives

$$\mathrm{OPT} - \mu(B_{\mathrm{ref}}) \le 2r\varepsilon.$$

On $\mathcal{E}_1^c$ we use $\Lambda \le r$. Therefore

$$\mathbb{E}[\Lambda] \le \sqrt{2}\, r \sqrt{\frac{a_T}{L_1}} + \frac{r}{2T^2}.$$

For the UCB phase,

$$\mathrm{OPT} - \mu(S_t) = \Lambda + \mathrm{OPT}_H - \mu(S_t).$$

It remains to bound the regret against the best basis inside H_{fin}. Let $N_t(e)$ and $\overline{\mu}_t(e)$ denote the count and empirical mean of arm e before UCB round t. The algorithm uses

$$U_t(e) = \overline{\mu}_t(e) + \sqrt{\frac{a_T}{2N_t(e)}}.$$

On the event

$$\mathcal{E}_2 = \left\{\forall e \in E,\ \forall s \le T : |\overline{\mu}_{e,s} - \mu_e| \le \sqrt{\frac{a_T}{2s}}\right\},$$

we have $\Pr(\mathcal{E}_2^c) \le 2nTe^{-a_T} \le 1/(2T)$. On $\mathcal{E}_2$, UCB optimism implies

$$\mathrm{OPT}_H - \mu(S_t) \le \sum_{e \in S_t} (U_t(e) - \mu_e) \le \sqrt{2a_T} \sum_{e \in S_t} \frac{1}{\sqrt{N_t(e)}}.$$

If q_e is the number of UCB rounds in which arm e is selected, then $\sum_{e \in H_{\mathrm{fin}}} q_e = rL_2$ and, because every final-memory arm has at least $L_1 \ge 1$ pre-UCB samples,

$$\sum_{t : e \in S_t} \frac{1}{\sqrt{N_t(e)}} \le \sum_{j=0}^{q_e - 1} (L_1 + j)^{-1/2} \le 2\left(\sqrt{L_1 + q_e} - \sqrt{L_1}\right).$$

Hence, by concavity of $q \mapsto \sqrt{L_1+q}-\sqrt{L_1}$ and $|H_{\text{fin}}| \le m$,

$$\sum_{t=1}^{L_2}(\text{OPT}_H - \mu(S_t)) \le 2\sqrt{2a_T}\sum_{e\in H_{\text{fin}}}\left(\sqrt{L_1+q_e}-\sqrt{L_1}\right) \le 2\sqrt{2a_T}\, m\left(\sqrt{L_1+\frac{rL_2}{m}}-\sqrt{L_1}\right).$$

The bad event contributes at most $rL_2 \Pr(\mathcal{E}_2^c) \le r/2$.

Combining the three terms gives

$$R_T \le r\kappa L_1 + \sqrt{2}\, rT\sqrt{\frac{a_T}{L_1}} + 2\sqrt{2a_T}\, m\left(\sqrt{L_1+\frac{rL_2}{m}}-\sqrt{L_1}\right) + r.$$

Let $x = (T^2 a_T/\kappa^2)^{1/3}$. Since $2^{-1/3}x \le L_1 \le 2^{-1/3}x + 1$,

$$r\kappa L_1 \le 2^{-1/3} r\kappa^{1/3} T^{2/3} a_T^{1/3} + r\kappa$$

and

$$\sqrt{2}\, rT\sqrt{\frac{a_T}{L_1}} \le 2^{2/3} r\kappa^{1/3} T^{2/3} a_T^{1/3}.$$

This proves the stated bound. Since $a_T = O(\log T)$ for fixed n, the bound is $o(T)$. □

Theorem 4.3 (Gap-dependent regret bound) *Assume that the optimal basis $B^\star$ is unique, and define the global basis gap*

$$\Delta_B = \min_{B\in\mathcal{B},\ B\neq B^\star}\big(\text{OPT} - \mu(B)\big) > 0.$$

Choose

$$L_1 = \left\lceil \frac{8a_T}{\Delta_B^2}\right\rceil, \qquad L_2 = T - \kappa L_1,$$

and assume T is large enough that $L_2 > 0$. Then Algorithm 1 satisfies

$$R_T \le r\kappa L_1 + r.$$

Consequently,

$$R_T = O\left(\frac{r\kappa \log(nT)}{\Delta_B^2}\right).$$

Proof: The streaming phase is unchanged from Theorem 4.2, and therefore contributes at most

$$R_{\text{stream}} \le r\kappa L_1.$$

We next show that the final memory contains $B^\star$ with high probability. The proof of Theorem 4.2 showed that after the stream ends, B_{ref} is an empirical maximum-weight basis of the full matroid. Let

$$\mathcal{E}_1 = \{\forall e \in E : |\widehat{\mu}_e - \mu_e| \le \varepsilon\}, \qquad \varepsilon = \sqrt{\frac{a_T}{2L_1}}.$$

By the choice of L_1, $\varepsilon \le \Delta_B/4$. Hence on $\mathcal{E}_1$, for every basis $B \neq B^\star$, Lemma 4.1 gives, with $d = |B^\star \setminus B|$,

$$\widehat{\mu}(B^\star) - \widehat{\mu}(B) \ge \mu(B^\star) - \mu(B) - 2d\varepsilon \ge d(\Delta_B - 2\varepsilon) > 0.$$

Thus $B^\star$ is the unique empirical maximum-weight basis, so $B_{\mathrm{ref}} = B^\star$ and $B^\star \subseteq H_{\mathrm{fin}}$. As before, Hoeffding's inequality and a union bound give

$$\Pr(\mathcal{E}_1^c) \le 2ne^{-a_T} \le \frac{1}{2T^2}.$$

It remains to analyze the UCB phase on the event that $B^\star$ is retained. Let

$$\mathcal{E}_2 = \left\{ \forall e \in E,\ \forall s \le T : |\overline{\mu}_{e,s} - \mu_e| \le \sqrt{\frac{a_T}{2s}} \right\}.$$

As in Theorem 4.2, $\Pr(\mathcal{E}_2^c) \le 2nTe^{-a_T} \le 1/(2T)$. We now work on $\mathcal{E}_1 \cap \mathcal{E}_2$. Since $B^\star \subseteq H_{\mathrm{fin}}$, the best basis in the final memory is $B^\star$. For a UCB round t, write

$$c_t(e) = \sqrt{\frac{a_T}{2N_t(e)}}.$$

Every arm in H_{fin} has at least L_1 samples before the UCB phase starts, and $N_t(e)$ only increases during the UCB phase. Therefore the choice of L_1 gives

$$c_t(e) \le \sqrt{\frac{a_T}{2L_1}} \le \frac{\Delta_B}{4} \qquad \text{for every } e \in H_{\mathrm{fin}}.$$

We claim that every UCB round chooses $B^\star$. Suppose instead that $S_t \ne B^\star$, and choose $e \in S_t \setminus B^\star$. By the strong basis exchange theorem, there exists $b \in B^\star \setminus S_t$ such that both $S_t - e + b$ and $B^\star - b + e$ are bases. Since $B^\star - b + e \ne B^\star$, the definition of Δ_B gives

$$\mu_b - \mu_e = \mu(B^\star) - \mu(B^\star - b + e) \ge \Delta_B.$$

On $\mathcal{E}_2$, $U_t(b) \ge \mu_b$ and $U_t(e) \le \mu_e + 2c_t(e) \le \mu_e + \Delta_B/2$. Hence

$$U_t(b) - U_t(e) \ge \mu_b - \mu_e - \frac{\Delta_B}{2} \ge \frac{\Delta_B}{2} > 0.$$

Replacing e by b strictly increases the UCB weight of S_t, while $S_t - e + b$ is feasible. This contradicts the definition of S_t as a maximum-UCB basis. Thus $S_t = B^\star$ on $\mathcal{E}_1 \cap \mathcal{E}_2$, and the UCB-phase regret is zero on this event.

On $(\mathcal{E}_1 \cap \mathcal{E}_2)^c$, the total UCB regret is at most rL_2. Thus the bad-event contribution is bounded by

$$rL_2 \Pr(\mathcal{E}_1^c) + rL_2 \Pr(\mathcal{E}_2^c) \le \frac{r}{2T} + \frac{r}{2} \le r.$$

Combining the streaming cost, the zero good-event UCB cost, and the bad-event term proves the stated finite-time bound. The order bound follows from $a_T = \Theta(\log(nT))$. □

5 Lower Bounds

The $T^{2/3}$ screening term in Theorem 4.2 is not an artifact of the analysis. This section adapts the single-pass streaming lower bound of Wang [20] to the present notation and then lifts it to rank-r matroids by a coupled direct-sum construction.

We first state the sample-memory primitive used in the lower bound. For $N \ge 1$ and $\Delta \in (0, 1/8]$, let $\mathsf{Trap}(N, \Delta)$ be the following distribution over N arms. An index I is drawn uniformly from $\{1, \ldots, N\}$; arm I has Bernoulli mean $1/2 + \Delta$, and all other arms have Bernoulli mean $1/2$.

Lemma 5.1 (Trapping a slightly better arm) *Consider a single pass over arms drawn from* $\mathsf{Trap}(N,\Delta)$. *If an algorithm uses fewer than* $N/(1200\Delta^2)$ *pulls from these* N *arms, then any set of at most* $N/8$ *stored arms contains the planted arm* I *with probability less than* $2/3$.

Proof: Let $Q = N/(1200\Delta^2)$, and let S be the set of arms stored by the algorithm after the pass, with $|S| \le N/8$. We prove the claim for an arbitrary randomized algorithm; the algorithm's internal randomness is included in the transcript and is independent of the instance.

For each $i \in \{1,\dots,N\}$, let $\mathbb{P}_i$ be the distribution of the full transcript when arm i is the planted arm. Let $\mathbb{P}_0$ be the null distribution in which every arm has Bernoulli mean $1/2$. Write τ_i for the number of pulls of arm i during the pass. By the chain rule for KL divergence under adaptive sampling,

$$\mathrm{KL}(\mathbb{P}_0,\mathbb{P}_i) = \mathbb{E}_0[\tau_i]\,\mathrm{kl}\left(\frac{1}{2},\frac{1}{2}+\Delta\right).$$

Indeed, the adaptive choices are deterministic functions of the past transcript and contribute no KL; only rewards observed from arm i have different laws under $\mathbb{P}_0$ and $\mathbb{P}_i$. Since $\Delta \le 1/8$,

$$\mathrm{kl}\left(\frac{1}{2},\frac{1}{2}+\Delta\right) = -\frac{1}{2}\log(1-4\Delta^2) \le 3\Delta^2.$$

The total number of pulls from the N arms is less than Q, hence

$$\sum_{i=1}^{N}\mathbb{E}_0[\tau_i] \le Q \quad \text{and therefore} \quad \frac{1}{N}\sum_{i=1}^{N}\mathrm{KL}(\mathbb{P}_0,\mathbb{P}_i) \le \frac{3\Delta^2 Q}{N} = \frac{1}{400}.$$

Now the planted index I is uniform over $\{1,\dots,N\}$. The probability that the final stored set captures the planted arm is

$$\Pr(I \in S) = \frac{1}{N}\sum_{i=1}^{N}\mathbb{P}_i(i \in S).$$

For every i,

$$\mathbb{P}_i(i \in S) \le \mathbb{P}_0(i \in S) + \|\mathbb{P}_i - \mathbb{P}_0\|_{\mathrm{TV}}.$$

Thus, using $|S| \le N/8$, Pinsker's inequality, and Jensen's inequality,

$$\begin{aligned}\Pr(I \in S) &\le \frac{1}{N}\mathbb{E}_0[|S|] + \frac{1}{N}\sum_{i=1}^{N}\|\mathbb{P}_i - \mathbb{P}_0\|_{\mathrm{TV}} \\ &\le \frac{1}{8} + \left(\frac{1}{2N}\sum_{i=1}^{N}\mathrm{KL}(\mathbb{P}_0,\mathbb{P}_i)\right)^{1/2} \le \frac{1}{8} + \frac{1}{\sqrt{800}} < \frac{2}{3}.\end{aligned}$$

This proves the lemma. □

This lemma is the sample-space tradeoff of Assadi and Wang [2], specialized to the constants needed here and used by Wang [20]. Intuitively, storing a small list that traps the unique Δ-better arm is already as hard as spending order N/Δ^2 samples across the stream.

Lemma 5.2 (High-confidence trapping under the null) *Let $N \geq 2$ and $0 < \Delta \leq 1/16$. Let $\mathbb{P}_0$ be the null instance on N arms, in which every arm has Bernoulli mean $1/2$. For each $i \in \{1, \dots, N\}$, let $\mathbb{P}_i$ be the planted instance in which arm i has Bernoulli mean $1/2 + \Delta$ and all other arms have Bernoulli mean $1/2$. Consider any adaptive algorithm that samples these arms and outputs a stored set $S \subseteq \{1, \dots, N\}$ with $|S| \leq N/8$ almost surely. Let τ_i be the number of pulls of arm i and let $Q = \sum_{i=1}^{N} \tau_i$.*

If, for some $0 < \alpha \leq 1/4$,

$$\frac{1}{N} \sum_{i=1}^{N} \mathbb{P}_i(i \in S) \geq 1 - \alpha,$$

then

$$\mathbb{E}_0 Q \geq \frac{N}{12\Delta^2} \log \frac{1}{\alpha}.$$

Equivalently, if

$$\bar{\alpha} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{P}_i(i \notin S),$$

then

$$\bar{\alpha} \geq \frac{1}{4} \exp\left(-12 \frac{\Delta^2 \mathbb{E}_0 Q}{N}\right).$$

Proof: Let $D_i = \{i \notin S\}$, and write

$$p_i = \mathbb{P}_0(D_i), \qquad q_i = \mathbb{P}_i(D_i).$$

Since $|S| \leq N/8$ almost surely, every transcript satisfies

$$\sum_{i=1}^{N} \mathbf{1}\{i \notin S\} = N - |S| \geq \frac{7N}{8}.$$

Taking expectation under $\mathbb{P}_0$ gives

$$\bar{p} := \frac{1}{N} \sum_{i=1}^{N} p_i = 1 - \frac{1}{N} \mathbb{E}_0[|S|] \geq \frac{7}{8}.$$

The trapping assumption gives

$$\bar{q} := \frac{1}{N} \sum_{i=1}^{N} q_i = \frac{1}{N} \sum_{i=1}^{N} \mathbb{P}_i(i \notin S) \leq \alpha.$$

Let $\mathrm{kl}(p, q)$ denote binary relative entropy. By data processing applied to the event D_i,

$$\mathrm{KL}(\mathbb{P}_0, \mathbb{P}_i) \geq \mathrm{kl}(p_i, q_i).$$

Using joint convexity of binary relative entropy,

$$\frac{1}{N} \sum_{i=1}^{N} \mathrm{KL}(\mathbb{P}_0, \mathbb{P}_i) \geq \frac{1}{N} \sum_{i=1}^{N} \mathrm{kl}(p_i, q_i) \geq \mathrm{kl}(\bar{p}, \bar{q}).$$

Since $\bar{p} \geq 7/8$, $\bar{q} \leq \alpha \leq 1/4$, and $\mathrm{kl}(p, q)$ is increasing in p and decreasing in q when $p > q$,

$$\mathrm{kl}(\bar{p}, \bar{q}) \geq \mathrm{kl}\left(\frac{7}{8}, \alpha\right).$$

Writing $L = \log(1/\alpha)$,

$$\mathrm{kl}\left(\frac{7}{8},\alpha\right) = \frac{7}{8}\log\frac{7/8}{\alpha} + \frac{1}{8}\log\frac{1/8}{1-\alpha}$$
$$= \frac{7}{8}L + \frac{7}{8}\log\frac{7}{8} + \frac{1}{8}\log\frac{1}{8(1-\alpha)}.$$

Because $\alpha \le 1/4$, $L \ge \log 4 > 1$. Also $\log(8/7) \le 1/7$ and $\log 8 \le 3$, while $1-\alpha \le 1$. Hence

$$\frac{7}{8}\log\frac{7}{8} + \frac{1}{8}\log\frac{1}{8(1-\alpha)} \ge -\frac{1}{2},$$

and therefore

$$\mathrm{kl}\left(\frac{7}{8},\alpha\right) \ge \frac{7}{8}L - \frac{1}{2} \ge \frac{1}{4}L.$$

Consequently,

$$\sum_{i=1}^{N} \mathrm{KL}(\mathbb{P}_0,\mathbb{P}_i) \ge \frac{N}{4}\log\frac{1}{\alpha}.$$

On the other hand, under $\mathbb{P}_0$ and $\mathbb{P}_i$ only rewards from arm i have different laws. The adaptive sampling decisions and the algorithm's internal randomness contribute no KL beyond the observed rewards, so the adaptive KL chain rule gives

$$\mathrm{KL}(\mathbb{P}_0,\mathbb{P}_i) = \mathbb{E}_0[\tau_i]\,\mathrm{kl}\left(\frac{1}{2},\frac{1}{2}+\Delta\right).$$

For $\Delta \le 1/16$,

$$\mathrm{kl}\left(\frac{1}{2},\frac{1}{2}+\Delta\right) = -\frac{1}{2}\log(1-4\Delta^2) \le 3\Delta^2.$$

Summing over i yields

$$\sum_{i=1}^{N} \mathrm{KL}(\mathbb{P}_0,\mathbb{P}_i) \le 3\Delta^2 \sum_{i=1}^{N} \mathbb{E}_0[\tau_i] = 3\Delta^2 \mathbb{E}_0 Q.$$

Combining the last two displays proves

$$\mathbb{E}_0 Q \ge \frac{N}{12\Delta^2}\log\frac{1}{\alpha}.$$

It remains only to derive the exponential form. If $\bar{\alpha} \le 1/4$, apply the implication just proved with $\alpha = \bar{\alpha}$ and rearrange to obtain

$$\bar{\alpha} \ge \exp\left(-12\,\frac{\Delta^2 \mathbb{E}_0 Q}{N}\right).$$

If $\bar{\alpha} > 1/4$, the claimed bound is trivial because the exponential factor is at most one. This proves the lemma. □

Lemma 5.2 is deliberately a null-instance statement: it lower bounds $\mathbb{E}_0 Q$. Turning it into a lower bound on the sampling effort under the planted mixture requires an additional change-of-measure or symmetrization argument and is not automatic.

Remark The constant $N/8$ in Lemma 5.2 is not special. For every fixed $\rho < 1$, the same proof gives a constant $c_\rho > 0$ such that the conclusion becomes

$$\mathbb{E}_0 Q \geq c_\rho \frac{N}{\Delta^2} \log \frac{1}{\alpha}$$

whenever the output set satisfies $|S| \leq \rho N$ almost surely. One only replaces the bound $\bar{p} \geq 7/8$ in the proof by $\bar{p} \geq 1-\rho$.

Theorem 5.3 (Rank-one streaming lower bound) *Assume $T \geq n-1$ and $m \leq (n-1)/20$. There is a universal constant $c > 0$ and a family of instances of the present model with $r = 1$ and $M = U_{1,n}$ such that every algorithm satisfying the single-pass memory constraint has*

$$\mathbb{E}[R_T] \geq c\,(n-1)^{1/3} T^{2/3}.$$

Proof: Let $K = n-1$ be the number of streaming arms. We prove the claim for even K; the odd case follows by ignoring one streaming arm and changing only universal constants. By Yao's principle, it is enough to give a distribution over instances on which every deterministic single-pass algorithm has expected regret $\Omega(K^{1/3}T^{2/3})$.

Set

$$\Delta = \frac{1}{8}\left(\frac{K}{T}\right)^{1/3}.$$

Since $T \geq K$, we have $\Delta \leq 1/8$. Let the initial arm be b_0 with Bernoulli mean $1/2$. The streaming arms are $e_1, \ldots, e_K$ and arrive in this order. The first half $e_1, \ldots, e_{K/2}$ is drawn from $\mathsf{Trap}(K/2, \Delta)$: one uniformly random arm e_I has mean $1/2 + \Delta$, and the other first-half arms have mean $1/2$. The arms $e_{K/2+1}, \ldots, e_{K-1}$ have mean $1/2$. Draw $Z \sim \text{Bernoulli}(1/2)$ independently of the first-half instance. If $Z = 1$, the last arm e_K has mean $3/4$; if $Z = 0$, it has mean $1/2$.

Because $M = U_{1,n}$, every basis is a single arm. Hence the regret of a play is the difference between the mean of the best arm in the full instance and the mean of the arm played. Let Q be the number of pulls made from the first-half streaming arms $e_1, \ldots, e_{K/2}$, and set

$$a = \frac{K}{2400\Delta^2}.$$

The random variable Q is independent of Z, because Z affects only the last arm, which arrives after the first half of the stream. We split the analysis into two cases.

First suppose

$$\mathbb{E}[Q] \geq \frac{a}{6}.$$

On the event $Z = 1$, the last arm e_K has mean $3/4$ and is the unique best arm. In that event, every pull from the first-half streaming arms has regret at least

$$\frac{3}{4} - \left(\frac{1}{2} + \Delta\right) = \frac{1}{4} - \Delta \geq \frac{1}{8}.$$

Therefore

$$\mathbb{E}[R_T] \geq \Pr(Z=1) \cdot \frac{1}{8}\,\mathbb{E}[Q] = \frac{a}{96} = \Omega(K^{1/3}T^{2/3}).$$

It remains to consider the case

$$\mathbb{E}[Q] < \frac{a}{6}.$$

Let C be the event that the planted arm e_I is stored after the first half of the stream. Truncate the algorithm so that, once its number of pulls from the first-half arms would reach a, it stops using further information from those arms and declares failure for capturing e_I. On the event $\{Q < a\}$, the truncated algorithm agrees with the original algorithm about whether e_I is stored. Since after the first half the algorithm stores at most $m \le K/20 < K/16 = (K/2)/8$ first-half arms, Lemma 5.1 gives

$$\Pr(C) \le \frac{2}{3} + \Pr(Q \ge a).$$

Thus, by Markov's inequality,

$$\Pr(C^c) \ge \frac{1}{3} - \Pr(Q \ge a) \ge \frac{1}{3} - \frac{\mathbb{E}[Q]}{a} > \frac{1}{6}.$$

On the event $Z = 0$, if C^c occurs, then the unique best arm in the full instance is e_I, with mean $1/2 + \Delta$, and once it is lost it can never be played again. Across all T rounds, at most Q pulls can be pulls of e_I. Therefore

$$R_T \ge \Delta(T\mathbf{1}\{C^c\} - Q) \qquad \text{on } Z = 0.$$

The event C^c and the random variable Q are independent of Z, so

$$\mathbb{E}[R_T \mid Z = 0] \ge \Delta(T\Pr(C^c) - \mathbb{E}[Q]) \ge \Delta\left(\frac{T}{6} - \frac{a}{6}\right).$$

The choice of Δ gives

$$\frac{a}{T} = \frac{64}{2400}\left(\frac{K}{T}\right)^{1/3} \le \frac{64}{2400} < \frac{1}{2},$$

where the last inequality uses $K \le T$. Therefore

$$\mathbb{E}[R_T] \ge \Pr(Z = 0)\,\mathbb{E}[R_T \mid Z = 0] \ge \frac{T\Delta}{24} = \Omega(K^{1/3}T^{2/3}).$$

In both cases the expected regret under the hard distribution is at least $cK^{1/3}T^{2/3}$ for a universal constant $c > 0$. Yao's principle then implies that every randomized algorithm has a fixed instance in this family with the same lower bound. Since $K = n - 1$, the theorem follows. □

To obtain the correct rank dependence, we use a coupled direct-sum construction. The hard prefixes in all blocks appear before any late arm, and a single random flag decides whether all late arms become very good. Thus excessive exploration of the prefixes is charged against the optimum in all r blocks at once, while insufficient exploration loses a constant fraction of the planted prefix arms.

Theorem 5.4 (Rank-r streaming lower bound) *Let $r \ge 1$ and let K be an even integer such that $K \ge 20(r+1)$. Set $n = r(K+1)$ and assume $T \ge rK$ and $r + 1 \le m \le K/20$. There is a universal constant $c > 0$ and a family of rank-r instances of the present model such that every algorithm satisfying the single-pass memory constraint has*

$$\mathbb{E}[R_T] \ge c\,r(rK)^{1/3}T^{2/3} = c\,r(n-r)^{1/3}T^{2/3}.$$

Proof: As before, it is enough by Yao's principle to give a distribution over instances on which every deterministic algorithm has the stated expected regret. Partition the ground set into r disjoint blocks $E_1, \dots, E_r$. In block j, let b_j be the initial arm, let

$$A_j = \{a_{j,1}, \dots, a_{j,K/2}\}, \qquad F_j = \{f_{j,1}, \dots, f_{j,K/2-1}\},$$

and let h_j be one late arm. Thus each block has K streaming arms. Let

$$M = \bigoplus_{j=1}^{r} U_{1,K+1}, \qquad B_0 = \{b_1, \dots, b_r\}.$$

A basis chooses exactly one arm from every block. All initial arms, all filler arms, and all non-planted prefix arms have Bernoulli mean $1/2$.

In each prefix A_j, independently choose a planted arm $g_j \in A_j$ uniformly at random and set its mean to $1/2 + \Delta$, where

$$\Delta = \frac{1}{8}\left(\frac{rK}{T}\right)^{1/3}.$$

Since $T \geq rK$, we have $\Delta \leq 1/8$. Finally draw one common random flag $Z \sim \text{Bernoulli}(1/2)$. If $Z = 1$, set $\mu_{h_j} = 3/4$ for every j; if $Z = 0$, set $\mu_{h_j} = 1/2$ for every j. The stream order is

$$A_1, A_2, \dots, A_r, F_1, \dots, F_r, h_1, \dots, h_r,$$

where each displayed set is listed in an arbitrary fixed internal order.

Fix a deterministic algorithm. For each block j, let τ_j be the number of pulls from arms in A_j before the end of the segment A_j, and let

$$Q = \sum_{j=1}^{r} \tau_j, \qquad a = \frac{K}{2400\Delta^2}.$$

The time intervals defining the τ_j's are disjoint. Moreover, during the segment A_j a basis contains only one arm from block j, so each global round contributes to at most one τ_j. The random variables $Q, \tau_1, \dots, \tau_r$ are independent of Z, because Z affects only the late arms, all of which arrive after the prefix segments.

First suppose

$$\mathbb{E}[Q] \geq \frac{ra}{6}.$$

On the event $Z = 1$, the optimal basis is $\{h_1, \dots, h_r\}$ and has mean $3r/4$. In every round counted by Q, no late arm has arrived, and the arm played in each block has mean at most $1/2 + \Delta$. Hence each such round has instantaneous regret at least

$$r\left(\frac{3}{4} - \frac{1}{2} - \Delta\right) = r\left(\frac{1}{4} - \Delta\right) \geq \frac{r}{8}.$$

Since the counted rounds are distinct and Q is independent of Z,

$$\mathbb{E}[R_T] \geq \Pr(Z = 1) \cdot \frac{r}{8}\,\mathbb{E}[Q] \geq \frac{r^2 a}{96} = \Omega\left(r(rK)^{1/3}T^{2/3}\right).$$

It remains to consider the case

$$\mathbb{E}[Q] < \frac{ra}{6}.$$

For each block j, let C_j be the event that g_j is stored at the end of the segment A_j. We claim that

$$\Pr(C_j) \le \frac{2}{3} + \Pr(\tau_j \ge a).$$

Indeed, condition on all information outside the random instance in A_j; it is only independent side information. Truncate the algorithm so that, once its number of pulls from A_j would reach the threshold a, it stops using further information from A_j and declares failure for capturing g_j. On the event $\{\tau_j < a\}$, this truncated algorithm agrees with the original algorithm about whether g_j is stored. At the end of A_j, the original algorithm stores at most $m \le K/20 < K/16 = (K/2)/8$ arms from A_j. Applying Lemma 5.1 with $N = K/2$ gives the displayed inequality.

Let

$$L = \sum_{j=1}^{r} \mathbf{1}\{C_j^c\}$$

be the number of planted prefix arms lost at the end of their prefix segments. Using Markov's inequality,

$$\begin{aligned} \mathbb{E}[L] &\ge \frac{r}{3} - \sum_{j=1}^{r} \Pr(\tau_j \ge a) \\ &\ge \frac{r}{3} - \frac{1}{a}\sum_{j=1}^{r} \mathbb{E}[\tau_j] = \frac{r}{3} - \frac{\mathbb{E}[Q]}{a} > \frac{r}{6}. \end{aligned}$$

Now condition on $Z = 0$. In each lost block j, the planted arm g_j is the unique best arm, with mean $1/2 + \Delta$, and once it is absent after A_j it can never be recovered. The algorithm can play g_j only during rounds counted by τ_j. Therefore

$$R_T \ge \Delta(TL - Q) \qquad \text{on } Z = 0.$$

Since L and Q are independent of Z,

$$\mathbb{E}[R_T \mid Z = 0] \ge \Delta\,(T\mathbb{E}[L] - \mathbb{E}[Q]) \ge \Delta\left(\frac{rT}{6} - \frac{ra}{6}\right).$$

Furthermore,

$$\frac{a}{T} = \frac{64}{2400}\left(\frac{K}{r^2 T}\right)^{1/3} \le \frac{64}{2400} < \frac{1}{2},$$

where we used $T \ge rK$ and $r \ge 1$. Hence

$$\mathbb{E}[R_T \mid Z = 0] \ge \frac{rT\Delta}{12}.$$

Multiplying by $\Pr(Z = 0) = 1/2$ gives

$$\mathbb{E}[R_T] \ge \frac{rT\Delta}{24} = \frac{1}{192}\, r(rK)^{1/3} T^{2/3}.$$

In both cases the hard distribution gives expected regret at least $c\, r(rK)^{1/3}T^{2/3}$ for a universal constant $c > 0$. Since $n - r = rK$, this is $c\, r(n-r)^{1/3}T^{2/3}$. Yao's principle converts the distributional lower bound for deterministic algorithms into a worst-case lower bound for randomized algorithms. $\square$

Since $\kappa = n - r + 1$, Theorem 5.4 matches the leading $r\kappa^{1/3}T^{2/3}$ dependence in Theorem 4.2, up to logarithmic factors and constants, in the stated small-memory parameter regime. The parity assumption on K only avoids floor and ceiling notation; odd K can be handled by ignoring one streaming arm per block and changing constants.

The same construction can be made gap-dependent by ensuring that both branches of the hard distribution have the same global basis gap. The only change is to replace each late arm by a pair of late arms whose means differ by the target gap. This keeps prefix exploration costly in one branch while making the planted prefix arm uniquely optimal, with the same gap, in the other branch.

Theorem 5.5 (Gap-dependent streaming lower bound) *Let $0 < \Delta \le 1/16$. Let $r \ge 1$ and let K be an even integer such that $K \ge 20(r+1)$. Set $n = r(K+1)$ and assume $T \ge rK$ and $r + 1 \le m \le K/20$. There is a universal constant $c > 0$ and a family of rank-r instances of the present model with unique optimal basis and global basis gap $\Delta_B = \Delta$ such that every algorithm satisfying the single-pass memory constraint has*

$$\mathbb{E}[R_T] \ge c\, r \min\left\{\frac{n-r}{\Delta^2}, \Delta T\right\}.$$

In particular, if $T \ge (n-r)/\Delta^3$, then

$$\mathbb{E}[R_T] \ge c\,\frac{r(n-r)}{\Delta^2}.$$

Proof: As in Theorem 5.4, it is enough to give a distribution over instances on which every deterministic algorithm has the stated expected regret. Partition the ground set into r disjoint blocks $E_1, \ldots, E_r$. In block j, let b_j be the initial arm, let

$$A_j = \{a_{j,1}, \ldots, a_{j,K/2}\}, \qquad F_j = \{f_{j,1}, \ldots, f_{j,K/2-2}\},$$

and let h_j^- and h_j^+ be two late arms. Thus each block has K streaming arms. Let

$$M = \bigoplus_{j=1}^{r} U_{1,K+1}, \qquad B_0 = \{b_1, \ldots, b_r\}.$$

All initial arms, filler arms, and non-planted prefix arms have Bernoulli mean $1/2$.

In each prefix A_j, independently choose a planted arm $g_j \in A_j$ uniformly at random and set its mean to $1/2 + \Delta$. Draw one common random flag $Z \sim \text{Bernoulli}(1/2)$. If $Z = 0$, set $\mu_{h_j^-} = \mu_{h_j^+} = 1/2$ for every j. If $Z = 1$, set

$$\mu_{h_j^+} = \frac{3}{4}, \qquad \mu_{h_j^-} = \frac{3}{4} - \Delta$$

for every j. The stream order is

$$A_1, \ldots, A_r, F_1, \ldots, F_r, h_1^-, h_1^+, \ldots, h_r^-, h_r^+,$$

where each displayed set is listed in an arbitrary fixed internal order.

Every instance in this distribution has a unique optimal basis and global basis gap exactly Δ. When $Z = 0$, the unique optimal basis is $\{g_1, \ldots, g_r\}$, and replacing any one g_j by a mean-$1/2$ arm loses exactly Δ. When $Z = 1$, the unique optimal basis is $\{h_1^+, \ldots, h_r^+\}$, and replacing any one h_j^+ by h_j^- loses exactly Δ; since $\Delta \le 1/16$, replacing h_j^+ by any prefix or filler arm loses more than Δ.

Fix a deterministic algorithm. For each block j, let τ_j be the number of pulls from arms in A_j before the end of the segment A_j, and let

$$Q = \sum_{j=1}^{r} \tau_j, \qquad a = \frac{K}{2400\Delta^2}.$$

The random variables $Q, \tau_1, \dots, \tau_r$ are independent of Z, because Z affects only the late arms, all of which arrive after the prefix segments. Set

$$q_\Delta = \frac{1}{6} \min\{ra, \Delta T\}.$$

First suppose $\mathbb{E}[Q] \ge q_\Delta$. On the event $Z = 1$, the optimal basis has mean $3r/4$. In every round counted by Q, no late arm has arrived, and the played arm in each block has mean at most $1/2 + \Delta$. Hence each such round has instantaneous regret at least $r(1/4 - \Delta) \ge 3r/16$. Therefore

$$\mathbb{E}[R_T] \ge \Pr(Z = 1) \cdot \frac{3r}{16} \mathbb{E}[Q] \ge \frac{3rq_\Delta}{32} = \Omega\left(r \min\left\{\frac{rK}{\Delta^2}, \Delta T\right\}\right).$$

It remains to consider the case $\mathbb{E}[Q] < q_\Delta$. For each block j, let C_j be the event that g_j is stored at the end of the segment A_j. The same truncation argument as in Theorem 5.4, together with Lemma 5.1 applied with $N = K/2$, gives

$$\Pr(C_j) \le \frac{2}{3} + \Pr(\tau_j \ge a).$$

Let

$$L = \sum_{j=1}^{r} \mathbf{1}\{C_j^c\}$$

be the number of planted prefix arms lost at the end of their prefix segments. Using Markov's inequality and the definition of q_Δ,

$$\begin{aligned} \mathbb{E}[L] &\ge \frac{r}{3} - \sum_{j=1}^{r} \Pr(\tau_j \ge a) \\ &\ge \frac{r}{3} - \frac{\mathbb{E}[Q]}{a} > \frac{r}{3} - \frac{q_\Delta}{a} \ge \frac{r}{6}. \end{aligned}$$

Now condition on $Z = 0$. In each lost block j, the planted arm g_j is the unique best arm, and once it is absent after A_j it can never be recovered. The algorithm can play g_j only during rounds counted by τ_j. Therefore

$$R_T \ge \Delta(TL - Q) \qquad \text{on } Z = 0.$$

Since L and Q are independent of Z,

$$\mathbb{E}[R_T \mid Z = 0] \ge \Delta(T\mathbb{E}[L] - \mathbb{E}[Q]) \ge \Delta\left(\frac{rT}{6} - q_\Delta\right).$$

If $\Delta T \le ra$, then $q_\Delta = \Delta T/6$, and the last display is at least $r\Delta T/16$. If $\Delta T > ra$, then $q_\Delta = ra/6$, and the last display is at least $r^2 a/16$. Multiplying by $\Pr(Z=0) = 1/2$, in both cases,

$$\mathbb{E}[R_T] = \Omega\left(r \min\left\{\frac{rK}{\Delta^2}, \Delta T\right\}\right).$$

Since $n - r = rK$, the theorem follows. □

We next record the logarithmic version of the preceding gap-dependent lower bound. It is asymptotic and uses the standard consistency condition from gap-dependent bandit lower bounds: an algorithm is *consistent* if, on every fixed instance θ in the model and for every $\gamma > 0$,

$$\mathbb{E}_\theta[R_T] = o(T^\gamma) \qquad \text{as } T \to \infty.$$

The finite-time lower bound above applies to arbitrary algorithms; consistency is what turns constant-confidence trapping into high-confidence trapping and therefore produces the logarithmic factor.

Theorem 5.6 (Logarithmic gap-dependent lower bound) *Let*

$$0 < \Delta \le \frac{1}{16}, \qquad r \ge 1, \qquad K \ge 20(r+1)$$

with K even. Set $n = r(K+1)$ and assume $r + 1 \le m \le K/20$. Then every consistent algorithm obeying the memory constraint has a rank-r instance with unique optimal basis and global basis gap $\Delta_B = \Delta$ such that

$$\liminf_{T\to\infty} \frac{\mathbb{E}[R_T]}{\log T} \ge c\,\frac{r(n-r)}{\Delta^2},$$

where $c > 0$ is a universal constant. Equivalently, for consistent algorithms,

$$\mathbb{E}[R_T] = \Omega\left(\frac{r(n-r)\log T}{\Delta_B^2}\right).$$

Proof: Let $N = K/2$. Use the same direct-sum partition matroid and stream order as in Theorem 5.5. Thus block j contains an initial arm b_j, a prefix

$$A_j = \{a_{j,1}, \dots, a_{j,N}\},$$

fillers, and two late arms h_j^-, h_j^+. The stream order is

$$A_1, \dots, A_r, F_1, \dots, F_r, h_1^-, h_1^+, \dots, h_r^-, h_r^+.$$

Define the null late-good instance θ^0 as follows. All initial, prefix, and filler arms have mean $1/2$, while in every block

$$\mu_{\theta^0}(h_j^+) = \frac{3}{4}, \qquad \mu_{\theta^0}(h_j^-) = \frac{3}{4} - \Delta.$$

The unique optimal basis is $\{h_1^+, \dots, h_r^+\}$, and its global basis gap is exactly Δ.

For each block j and prefix index $i \in \{1, \dots, N\}$, define an alternative instance $\theta^{j,i}$ that differs from θ^0 only inside block j before the late arms can affect the transcript. In block j, set

$$\mu_{\theta^{j,i}}(a_{j,i}) = \frac{1}{2} + \Delta$$

and set every other arm in that block, including h_j^- and h_j^+, to have mean $1/2$. All other blocks are as in θ^0. Then $\theta^{j,i}$ has the unique optimal basis

$$B_{j,i}^\star = \{a_{j,i}\} \cup \{h_\ell^+ : \ell \neq j\},$$

and again $\Delta_B = \Delta$: replacing $a_{j,i}$ in block j loses exactly Δ, replacing h_ℓ^+ by h_ℓ^- in another block loses exactly Δ, and replacing h_ℓ^+ by a prefix, filler, or initial arm loses at least $1/4 > \Delta$.

For each block j, define the random set

$$S_j = S_j^{\text{store}} \cup S_j^{\text{heavy}},$$

where S_j^{store} is the set of arms from A_j stored at the end of segment A_j, and S_j^{heavy} is the set of arms from A_j that are pulled at least $T/2$ times before the end of segment A_j. This set records whether a prefix arm has either been retained or already played heavily before it can be lost. Its size is always at most $m+2$: at most m arms from A_j can be stored, and at most two arms can each receive $T/2$ pulls during the horizon. Since $m \leq K/20 = N/10$ and $N = K/2 \geq 20$, we have

$$|S_j| \leq m + 2 \leq N/5.$$

On instance $\theta^{j,i}$, if $a_{j,i} \notin S_j$, then $a_{j,i}$ is not stored at the end of A_j and was pulled fewer than $T/2$ times before that point. It cannot be recovered after leaving A_j, so the algorithm plays the unique optimal arm in block j fewer than $T/2$ times over the whole horizon. Every other arm in block j has mean $1/2$, hence

$$R_T \geq \frac{\Delta T}{2} \qquad \text{on } \{a_{j,i} \notin S_j\}.$$

By consistency, for every fixed $\eta \in (0,1)$ and all sufficiently large T,

$$\mathbb{P}_{\theta^{j,i}}(a_{j,i} \notin S_j) \leq \frac{2\mathbb{E}_{\theta^{j,i}}[R_T]}{\Delta T} \leq T^{-1+\eta}.$$

Hence, with

$$\alpha_{j,T} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{P}_{\theta^{j,i}}(a_{j,i} \notin S_j),$$

we have $\alpha_{j,T} \leq T^{-1+\eta}$ for all large T.

Now apply the constant-fraction version of Lemma 5.2 from the preceding remark to segment A_j, with output set S_j. The transcript on A_j under θ^0 is exactly the null transcript in the lemma, while the alternatives $\theta^{j,i}$ are the planted instances up to the end of A_j. If τ_j is the number of pulls from arms in A_j before the end of segment A_j under θ^0, then for some universal constant $c_0 > 0$,

$$\mathbb{E}_{\theta^0}\tau_j \geq c_0 \frac{N}{\Delta^2} \log \frac{1}{\alpha_{j,T}} \geq c_0(1-\eta)\frac{N}{\Delta^2}\log T.$$

Summing over blocks and writing $Q = \sum_{j=1}^{r} \tau_j$,

$$\mathbb{E}_{\theta^0} Q \geq c_0(1-\eta)\frac{rN}{\Delta^2}\log T = \frac{c_0(1-\eta)(n-r)}{2\Delta^2}\log T.$$

On θ^0, every round counted by Q occurs before the late arms arrive. The optimal basis has mean $3r/4$, whereas every playable arm in each block has mean at most $1/2$ during those prefix segments. Each counted round therefore incurs instantaneous regret at least $r/4$. Since the prefix segments are disjoint, Q counts distinct rounds, and

$$\mathbb{E}_{\theta^0}[R_T] \geq \frac{r}{4}\,\mathbb{E}_{\theta^0}Q \geq \frac{c_0(1-\eta)}{8}\,\frac{r(n-r)}{\Delta^2}\log T.$$

Taking $\liminf_{T\to\infty}$ and then letting $\eta \downarrow 0$ proves the claim with a universal constant. $\square$

Optimizing Theorem 5.5 over $\Delta \asymp ((n-r)/T)^{1/3}$ recovers the gap-free lower bound in Theorem 5.4, up to constants. For a fixed positive gap and long enough horizon, it gives an unavoidable screening cost of order $r(n-r)/\Delta_B^2$, matching the screening term in Theorem 4.3 up to logarithmic factors and the difference between $n-r$ and $\kappa = n-r+1$. The logarithmic Theorem 5.6 shows that this logarithmic factor is also necessary for consistent algorithms. Since, for fixed n, $\log(nT)$ and $\log T$ are equivalent up to constants, Theorem 4.3 and Theorem 5.6 match in the dependence on rank, stream length, horizon, and the global basis gap. For the gap-dependent choice of L_1, every retained arm is already sampled enough during screening to separate all basis exchanges by the global basis gap, so the final UCB phase incurs regret only on the low-probability failure event.

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